

Finding the value of an algebraic expression

Substituting a number for the letter — including a negative one, where the brackets matter.

LESSON 13–15

3 × 40 MIN

MATEMATIKA 6, §3

S7 2.2

LESSON CLOCK

20'

25'

30'

30'

15'

Substituting a number	20 min
Negative values	25 min
Using formulae	30 min
Tables of values	30 min
Homework	15 min

LEARNING OBJECTIVES

- Substitute a value and evaluate, respecting the order of operations.
- Substitute a negative value using brackets.
- Use a formula to compute a required quantity.
- Build a table of values for an expression.

TEXTBOOK

National	pp. 35–42
Cambridge	Stage 7, pp. 24–28
Workbook	Exercise 2.2

01

Explanation

Substituting a number

Replace the letter by the number, then follow the order of operations exactly as in arithmetic.

EXPRESSION	AT $X = 4$	VALUE
$3x + 2$	$3 \cdot 4 + 2$	14
$3(x + 2)$	$3 \cdot 6$	18
x^2	$4 \cdot 4$	16
$2x^2$	$2 \cdot 16$	32
$(2x)^2$	8^2	64

2X² AND (2X)² ARE DIFFERENT

In $2x^2$ only x is squared; in $(2x)^2$ the whole product is. Powers act before multiplication unless a bracket says otherwise.

Negative values

Substitute a negative number **inside brackets**, and the sign rules do the rest.

EXPRESSION	AT $x = -3$	VALUE
$2x + 5$	$2 \cdot (-3) + 5$	-1
x^2	$(-3)^2$	9
$-x$	$-(-3)$	3
$x^2 - x$	$9 - (-3)$	12
$5 - 2x$	$5 - 2(-3)$	11

-x IS NOT ALWAYS NEGATIVE

It means “the opposite of x ”. When x is negative, $-x$ is positive. Reading the minus sign as “opposite” rather than “negative” removes the confusion entirely.

Using formulae

FORMULA	DATA	VALUE
$P = 2(a + b)$	$a = 7, b = 4$	22 cm
$S = ab$	$a = 7, b = 4$	28 cm^2
$s = vt$	$v = 60, t = 3$	180 km
$S = \frac{1}{2}ah$	$a = 10, h = 6$	30 cm^2
$C = \frac{5}{9}(F - 32)$	$F = 68$	$20\text{ }^\circ\text{C}$

CARRY THE UNITS THROUGH

A formula answers with a quantity, not a bare number: 22 cm , 28 cm^2 , 180 km . The units also check the work — an area that comes out in cm means something went wrong.

Tables of values

A table shows how the value moves as the letter changes — the first step towards a graph.

X	-2	-1	0	1	2	3
$2x + 1$	-3	-1	1	3	5	7
x^2	4	1	0	1	4	9
$10 - 3x$	16	13	10	7	4	1

TWO PATTERNS TO NOTICE

$2x + 1$ climbs by 2 each step and $10 - 3x$ falls by 3, while x^2 does neither — it falls, reaches zero and climbs again. Grade 7 will call the first two *linear*.

02

Worked examples

EXAMPLE 1 Find the value of $x^2 - x$ at $x = -3$.

1

Substitute in brackets: $(-3)^2 - (-3)$.

2

 $(-3)^2 = 9$

3

 $-(-3) = +3$

4

 $9 + 3 = 12$

ANSWER

12

EXAMPLE 2 A car travels at 60 km/h for 3 hours. Use $s = vt$.

① $s = 60 \cdot 3$

② $= 180$

③ With units: 180 km.

ANSWER

180 km

EXAMPLE 3 Convert 68 °F using $C = \frac{5}{9}(F - 32)$.

① $F - 32 = 36$
Bracket first.

② $\frac{5}{9} \cdot 36$

③ $= 20$

④ 20 °C.

ANSWER

20 °C

03 Interactive model

Insist on brackets round every substituted negative number; the sign errors that plague the whole year begin here and end here.

Substitute into $2x + 1$

$f(x) = 2x + 1$

x 2

denominator 1

value 5

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To substitute	O‘rniga qo‘yish	Подставить
Value of an expression	Ifodaning qiymati	Значение выражения
Formula	Formula	Формула
Table of values	Qiymatlar jadvali	Таблица значений
To evaluate	Hisoblash	Вычислить
Order of operations	Amallar tartibi	Порядок действий
Square	Kvadrat	Квадрат
Units	O‘lchov birligi	Единицы измерения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$3x + 2$ at $x = 4$ is:

14

18

20

12

$3(x + 2)$ at $x = 4$ is:

14

18

20

12

x^2 at $x = -3$ is:

9

-9

6

-6

$-x$ at $x = -3$ is:

$(2x)^2$ at $x = 4$ is:

$s = vt$ with $v = 60$, $t = 3$ gives:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3x + 2$ at $x = 4$

ANSWER 14

2. $3(x + 2)$ at $x = 4$

ANSWER 18

3. x^2 at $x = 4$

ANSWER 16

4. $2x^2$ at $x = 4$

ANSWER 32

5. $(2x)^2$ at $x = 4$

ANSWER 64

6. $2x + 5$ at $x = -3$

ANSWER -1

7. $-x$ at $x = -3$

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. $x^2 - x$ at $x = -3$

ANSWER 12

2. $5 - 2x$ at $x = -3$

ANSWER 11

3. $P = 2(a + b)$ at $a = 7, b = 4$

ANSWER 22 cm

4. $S = ab$ at $a = 7, b = 4$

ANSWER 28 cm²

5. $S = \frac{1}{2}ah$ at $a = 10, h = 6$

ANSWER 30 cm²

6. $s = vt$ at $v = 60, t = 3$

ANSWER 180 km

7. A table of $2x + 1$ at $x = -1, 0, 1$

ANSWER -1, 1, 3

Hard · stretch and reason

7 PROBLEMS

1. $C = \frac{5}{9}(F - 32)$ at $F = 68$

ANSWER 20°C

2. $C = \frac{5}{9}(F - 32)$ at $F = 32$

ANSWER 0°C

3. $x^2 + 2x - 3$ at $x = -4$

ANSWER 5

4. $(x - 1)(x + 2)$ at $x = -2$

ANSWER 0

5. $\frac{a + b}{2}$ at $a = -7, b = 3$

ANSWER -2

6. For which x does $2x + 1$ equal 9?

ANSWER $x = 4$

7. Why is $2x^2$ not $(2x)^2$?

ANSWER The power acts on x alone unless a bracket says otherwise

07 Homework

Homework — 5 tasks

Put every negative substitution in brackets, without exception.

- Find $4x - 3$ at $x = 5$ and at $x = -5$.
- Find $x^2 + x$ at $x = -4$.
- Find the perimeter and area of a rectangle with $a = 9$ and $b = 6$.
- Convert 86°F to Celsius.

5. Make a table of $10 - 3x$ for $x = -2$ to 3.